



MPAIRWE LAUBEN

DEVSECOPS • ML • AGENTIC AI
ENGINEER

ABOUT

DevSecOps engineer and applied ML practitioner. Co-founder of IONATECH (ionatec.com), an AI startup, and engineering partner at Sauti Health (sautihealth.com), building emergency-dispatch infrastructure for sub-Saharan Africa. By day, CI/CD, cloud, and security posture at AIBOS Uganda. Outside: training models, building agentic systems on LangChain and LangGraph, and instrumenting MCP servers that expose tools to LLMs.

Kampala,
Uganda

+256 773 336
896

✉ mpairwelauben75@gmail.com

WORK EXPERIENCE

AIBOS Uganda

Kampala

Feb 2024 – Present

DEVSECOPS ENGINEER

- ▶ Own CI/CD posture on GitHub Actions, cutting deploy time ~60% while adding SBOM generation, container scanning, and signature verification
- ▶ Run infrastructure-as-code (Terraform) across AWS, Azure, and GCP with policy-as-code gates on every plan
- ▶ Harden Kubernetes workloads with Pod Security Standards, NetworkPolicies, and supply-chain controls (cosign / Trivy)
- ▶ Wire Prometheus + Grafana + Loki dashboards with SLO-driven alerts and on-call runbooks
- ▶ Drive compliance posture (encryption, RBAC, secrets via Vault) across the cloud estate

GITHUB ACTIONS • TERRAFORM •
KUBERNETES • TRIVY • COSIGN • VAULT •
PROMETHEUS • GRAFANA • AWS

Sauti Health

Kampala

2026 — Present

DEVELOPER · ENGINEERING PARTNER

- ▶ Partner with the Sauti Health team on the engineering behind a pre-hospital emergency dispatch platform for sub-Saharan Africa
- ▶ Build and harden backend services that coordinate community responders and motorcycle riders over USSD on basic 2G phones
- ▶ Ship offline-first responder workflows on low-end Android, with sync, retries, and conflict resolution for patchy networks
- ▶ Integrate mobile-money payment flows (MTN MoMo · Airtel Money) and DHIS2 reporting for health-system interoperability
- ▶ Instrument telemetry, dashboards, and CI gates on safety-critical paths so dispatch reliability stays measurable

PYTHON · FASTAPI · POSTGRESQL · USSD/IVR · MOBILE MONEY · DHIS2 · OFFLINE-FIRST · OBSERVABILITY

IONATECH

Kampala

Feb 2023 — Present

CO-FOUNDER · ML & AGENTIC ENGINEER

- ▶ Co-founded IONATECH (ionatec.com) — a startup building applied ML, agentic AI, and secure cloud systems for East African operators
- ▶ Ship LLM agents with LangChain + LangGraph: tool-use, retrieval, multi-agent orchestration, and evals
- ▶ Build RAG pipelines on Qdrant / pgvector with hybrid retrieval, rerank, and Braintrust/ Promptfoo eval harnesses
- ▶ Deliver classical ML for classification, regression, and ranking — with monitoring, drift detection, and retraining loops
- ▶ Run the engineering curriculum and mentor residents from prototype through production

LANGCHAIN · LANGGRAPH · MCP · QDRANT · BRAINTRUST · PYTHON · FASTAPI · NEXT.JS · SCIKIT-LEARN

VERCEL
KUBERNETES
DOCKER
HELM
PROMETHEUS
GRAFANA
LOKI
OPENTELEMETRY
ISTIO
POSTGRESQL
PGVECTOR
MYSQL
MONGODB
REDIS
QDRANT
SNOWFLAKE
FIREBASE
SUPABASE
NEXT.JS
REACT
TYPESCRIPT
TAILWIND
SHADCN/UI
MOTION
NODE.JS
FASTAPI
SPRING BOOT
TRPC
FLUTTER
REACT NATIVE
KOTLIN
SWIFT
FIREBASE
APP STORE / PLAY

LINKS

-  LINKEDIN
mpairwe-lauben-3a1461344
-  GITHUB
github.com/mpairweLandwind
-  IONATECH
ionatec.com
-  SAUTI HEALTH
sautihealth.com

Makerere University
Kampala
Jan 2022 – Present

PAHAPPA LIMITED
Kampala
Jun 2022 – Jan 2024

WEB INFO-NET LIMITED
Kampala
Jun 2018 – Aug 2022

STUDENT SOFTWARE ENGINEER

- ▶ Built TaskApp — a Kotlin task management app with real-time Firebase sync
- ▶ Developed scalable cross-platform apps using React Native and Flutter
- ▶ Implemented secure JWT and biometric authentication flows
- ▶ Delivered a fintech mobile app with digital payments, rewards, and transaction tracking
- ▶ Integrated third-party payment APIs and designed scalable backend architecture

KOTLIN · REACT NATIVE · FLUTTER · FIREBASE · JWT · FINTECH

DEVOPS INTERN → ENGINEER

- ▶ Built enterprise applications with Spring Boot (MVC, Data JPA, Security, Cloud)
- ▶ Used Hibernate ORM/JPA with MySQL, PostgreSQL, Oracle, and MongoDB databases
- ▶ Designed Microservice and Monolithic architectures based on project requirements
- ▶ Developed RESTful APIs and event-driven systems for high-throughput workflows

SPRING BOOT · JAVA · HIBERNATE · MYSQL · POSTGRESQL · MICROSERVICES · REST API

APPLICATION DEVELOPER INTERN & TRAINER

- ▶ Developed mobile apps using Swift, Java, and Kotlin for iOS and Android platforms
- ▶ Built an Odoo-based Employment Management System for HR automation
- ▶ Implemented secure authentication and third-party API integrations
- ▶ Provided Level 1 technical support, troubleshooting, and end-user training

SWIFT · JAVA · KOTLIN · OD00 · IOS · ANDROID

LANGUAGES

English (Fluent)
German (Basic)

PERSONAL DETAILS

DATE OF BIRTH
22 October 1995

NATIONALITY
Ugandan

LOCATION
Kampala, Uganda

REFERENCES

Dr. Kimbugwe Nasser
Lecturer & Academic Supervisor
Makerere University, School of Computing & IT

Kalema Martin
Supervisor
AIBOS Uganda

Dr. Mark Magumba
Senior Lecturer
Makerere University, School of Computing & IT

Fahadi Nguyenguye
Technical Lead
PAHAPPA LIMITED

Dr. Ggaliwango Marvin
Lecturer & AI/ML Mentor
Makerere University, School of Computing & IT

ZALONIDENTALHUB
Kampala
Apr 2022 – Present

- LEADING DEVELOPER & MAINTAINER**
- ▶ Led full SDLC for dental practice management software from conception to delivery
 - ▶ Built C#/VB.NET desktop systems for inventory and patient management
 - ▶ Developed a Flutter mobile app with e-commerce, appointments, and booking features
 - ▶ Ensured seamless desktop-mobile data interoperability
 - ▶ Led a team of developers, defining software architectures and coding standards

C# · VB.NET · FLUTTER · DESKTOP · MOBILE · TEAM LEAD

EDUCATION

Makerere University
2022 – Expected 2027
CGPA 4.31

Bachelor of Science in Software Engineering

- ▶ Specialising in AI/ML and Enterprise Application Development
- ▶ Integrated Supabase, MongoDB, Cloudflare, and Firebase for scalable storage
- ▶ Implemented RBAC and encryption for secure multi-role systems
- ▶ Built a full HRMS system with advanced role and permissions management
- ▶ Worked on IoT and Embedded Systems research projects
- ▶ Optimised PostgreSQL queries for high-performance data retrieval

Makerere University
2019 – 2023

Bachelor of Science in Computer Engineering

- ▶ Transitioned to the Software Engineering programme
- ▶ Studied Digital Electronics and hardware design principles
- ▶ Hardware repair, maintenance, and systems diagnostics

**Uganda Technical
College – Lira**

2016 – 2018

CGPA 4.77

**National Diploma in Information
Communication Technology**

- ▶ Database Management and Web Development
- ▶ Internet technologies and networking fundamentals
- ▶ RBAC and Encryption implementation
- ▶ Java programming and object-oriented design
- ▶ Networking and Server Configuration

**Boston High School –
Mpala**

2011 – 2015

**Uganda Certificate of Education
(UCE)**

- ▶ Graduated in Top 5% of class
- ▶ Varsity Debate Team Captain

CERTIFICATIONS

PLANNED	AWS Certified DevOps Engineer · Professional	— Amazon Web Services
PLANNED	Certified Kubernetes Security Specialist	— CNCF
PLANNED	Certified Kubernetes Administrator	— CNCF
PLANNED	HashiCorp Certified · Terraform Associate	— HashiCorp
PLANNED	AWS Solutions Architect · Associate	— Amazon Web Services
PLANNED	Google Cloud Professional ML Engineer	— Google Cloud